

cuit. You can simulate a large number of analogue components from LTC as well as other discrete passive devices contained in the library. Third-party models, like MOSFET components, can be imported to the workspace and integrated in your schematic; their configuration can be displayed without using internal nodes and without affecting the switching waveforms.

It is possible to configure each component in the schematic individually by right-clicking on it. This way the user can modify the functions and parameters. For example, the resistance, tolerance level and power rating of a resistor, parasitic properties and the amplitude of a voltage source can be changed.

Further, the general-purpose schematic and symbol-capture and editing tools can help the user insert all kinds of geometrical figures and shapes found in complex output designs. It supports unlimited schematic size and hierarchy. You can also generate netlists and cross probe simulation data.

Advanced options

The enhancements to Spice makes the simulation of SMPS very fast and easy compared to typical Spice simulators. This allows you to view and evaluate waveforms for most switching regulators in a few minutes compared to hours required by other similar Spice simulators.

Once the schematic is created, you can evaluate the validity of the scheme with the help of built-in compiler and simulator. LTSpice provides you with advanced simulation and analyses options that can outperform some paid Spice simulators. It has features like steady-state detection, step response computation and Fourier analysis. You can also plot circuit simulations

based on transient, AC, noise and DC analysis.

LTSpice is capable of computing bill of materials (for certain circuits), heat dissipation of components and generate complete efficiency reports of the SMPS circuits you have drafted. This lets you determine the points with higher energy dissipation and thereby take measures to minimise the energy wastage in the circuit.

Easy to use

A user with no previous knowledge about the device, Spice or schematic capture can get a detailed simulation of power systems with just a few mouse clicks. A good collection of pre-drafted and verified demo circuits can be used to start building the custom circuit, and the parameters can be suitably modified to fit different power supply requirements. All these features make LTSpice easy to understand and operate.

Installation and getting started

This download includes fully-functional LTSpice IV, help files, over 1100 macro models for 80% of LT's power products, numerous operational amplifiers, resistors, transistors and MOSFETs. The installation file LTSpiceIV.exe is a self-extracting gzip compressed file. Double clicking on it installs LTSpice IV in the computer as it extracts. Windows Vista users should right click on file and select 'Run as administrator' option.

Hundreds of demo circuits are available for download at Linear Technology website, all of which were designed and reviewed by the firm's factory applications group.

The pre-drafted test fixtures (JIG) are used for testing models during

LTSpice IV Facts

License type	Freeware
Developer	Mike Engelhardt, Linear Technology Corporation
Latest version	4.21x
Operating systems and requirements	➤ Windows - Version: Windows 2000 and later (Runs on both 32-bit and 64-bit operating system) - Processor: Pentium 4 ➤ Linux - Via Wine Windows simulator ➤ OS X - Via Crossover, Parallels, or DARWINE

development and provide a good starting point for draft. You can also create your own design using the schematic editor with the help of a large number of macro models of power devices provided in the software package.

Support

The software is updated often. You can check for updates and install them by built-in menu command *Tools > Sync Release*. You have to close all other windows and establish an Internet connection before the update can be made.

Comments about the software and bug report can be directly mailed to LTSpice@linear.com. By registering with MyLinear account at www.linear.com, you would be able to receive free samples, literature, product updates and so on. Blogposts and video tutorials for different functions that can be done using this tool are available in the website (Design Support > Design Simulation Page > LTSpice IV > LTSpice Blog). For additional support you could also join the Yahoo group LTSpice IV and LTSpice wiki www.ltwiki.org, which are not affiliated to Linear Technology. ●

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